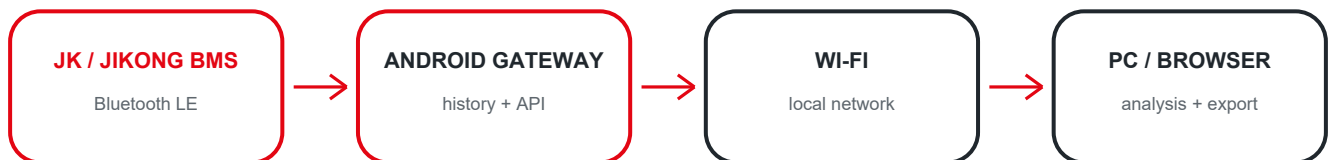


COMPLETE USER MANUAL

BMS Data Platform

Local battery monitoring and diagnostics for JK/Jikong BMS

Manual release
Desktop Monitor 0.7.35 · Android Gateway 0.4.9 · 18 September 2026



Purpose

The application turns an Android phone into a local BMS data gateway and a computer or browser into a complete workspace for long-term monitoring, charting, anomaly discovery and history export. No cloud account is required.

Normal monitoring is read-only. The only controlled exception is the separately armed experimental pulse-resistance test, which briefly toggles the Charge MOS and never changes BMS protection thresholds.

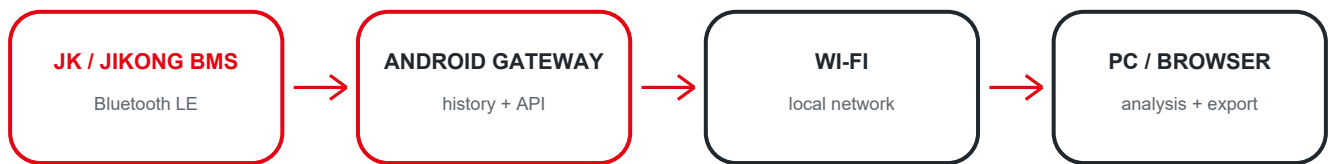
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How to use this manual

For a new installation, follow chapters 3–5 in order. If live data already reaches the PC, start with chapters 6–11. Reliable diagnostics require accumulated history and comparisons at similar SOC, current and temperature.

1. System architecture and operating principle



Component responsibilities

JK/Jikong BMS	Telemetry source: cell voltages, current, SOC, temperatures, MOS states, balancing, alarms and supported configuration values.
Android gateway	Connects to one selected BMS over Bluetooth LE, displays key values, stores timestamped history and exposes a local HTTP API over Wi-Fi.
Desktop monitor	Downloads live data and missing history segments, renders charts, runs comparative diagnostics and exports data.
Browser	Opens the same desktop interface through the PC address. The PC must remain on and the browser device must be on the same network.

What the system does

- shows live telemetry and clearly marks stale data after a connection loss;
- records history on the phone and synchronizes only missing intervals to the PC;
- renders standalone and combined charts, connection gaps and SOC events;
- compares cells at similar load, SOC and temperature;
- exports Excel and SQL/SQLite data for external analysis;
- provides a JSON API for Zabbix, Home Assistant and custom scripts.

What the system does not do

- it does not replace BMS hardware protection or certified measuring equipment;
- it cannot infer exact SOC from voltage alone; the BMS SOC may drift;
- normal monitoring never changes BMS protection settings;
- it does not automatically upload data to a cloud service or create Internet remote access.

2. Compatibility, requirements and safety

Supported BMS	JK/Jikong devices using JK02 and JK04 protocol variants. Practical compatibility depends on model and firmware. The architecture targets roughly 8S–32S systems and dynamically follows the received cell count.
Android	Android 8.0 or later. Android 12+ uses the Nearby devices permission. Android 8–11 normally needs Location permission and the system Location switch enabled; this is an Android BLE scanning rule, not user tracking.
Windows	64-bit Windows. Port 4174 must be available. Windows Firewall may ask to allow private-network access.
Network	Phone and PC on the same trusted Wi-Fi. Default gateway port: 8765. Monitor port: 4174.
Languages	Fully supported user interfaces: English, Ukrainian and Russian.

Safety

Use the software only with a correctly configured BMS and safe charging hardware. Never use a card color or software alert as the only protection. Verify BMS thresholds, fusing, wiring and temperature sensors independently.

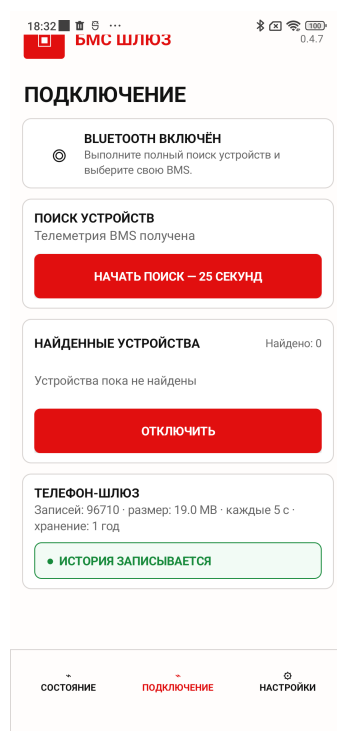
Local-network privacy

The phone API has no account, HTTPS or built-in authentication. That is convenient on a trusted home LAN, but any device on that network may be able to read telemetry. Do not expose port 8765 directly to the Internet. Use a trusted VPN or protected gateway for remote access.

Signing and updates

Download the APK and Windows archive only from the official release. Future APKs must use the same release signing key so Android can update the installed gateway without losing local data. Do not uninstall the gateway before an update if its settings and history must be preserved.

3. Installing the Android gateway



Real Android gateway screen. The image shows a closely related point release; the current gateway version is 0.4.9.

Step-by-step installation

- Open the official release at <https://github.com/zayants/bms-data-platform/releases/tag/v0.7.35> and download BMS-Gateway-0.4.9-Android.apk.
- Allow your browser or file manager to install from that source. This is an Android source permission; disabling Play Protect is not required.
- Run the APK and install over the existing app during an update. If Android reports an incompatible signature, stop and verify the source.
- Open the app and grant the requested Bluetooth permissions. On Android 8–11, also grant Location and enable the system Location switch.
- Exclude the gateway from battery optimization if the phone vendor aggressively stops background apps.

Play Protect

A normal scan of an unknown APK is expected. If Android says “Harmful app blocked”, do not instruct users to disable protection. Verify the download source, signing identity and release SHA-256 instead.

Why keep the phone near the battery

BLE communication is normally more stable at short range. An old phone can be mounted near the battery, supplied from a safe permanent power source, and used as a status screen, history recorder and Wi-Fi gateway.

4. Discovering and connecting the BMS

Use the Connection page shown in the preceding illustration. It runs a complete 25-second scan, lists discovered devices and reports the history-recording state.

Correct connection sequence

- Enable Bluetooth. On Android 8–11, enable Location as well; otherwise Android may return an empty BLE scan.
- Close the official JK BMS app. A BMS commonly accepts only one active BLE client.
- Tap “Start scan — 25 seconds” and wait for the countdown to finish. The gateway lists all discovered BLE devices, not only hard-coded JK names.
- Move closer if the signal is weak. Select the BMS by the name or address known to you.
- Wait for “BMS telemetry received”. Confirm that the cell count, pack voltage and current are plausible.

Automatic reconnect

The gateway remembers the selected BMS and attempts to restore it after a short interruption. A successful connection must not be repeatedly torn down by background scans. To use another BMS, start a manual scan and explicitly select the new device.

If the BMS beeps repeatedly

This usually indicates repeated failed connection attempts. Close competing BMS apps, cycle Bluetooth once, move closer, verify permissions, then run one complete scan. Do not hammer the scan button.

Plausibility checks

- pack voltage should approximately equal the sum of active cell voltages;
- current sign must match the operating mode: positive is charge, negative is discharge;
- the last-packet age should update every one or two seconds;
- when the BMS is off, data must become stale and charts must show a gap instead of pretending old values are current.

5. Wi-Fi gateway and desktop monitor

Enable the phone gateway

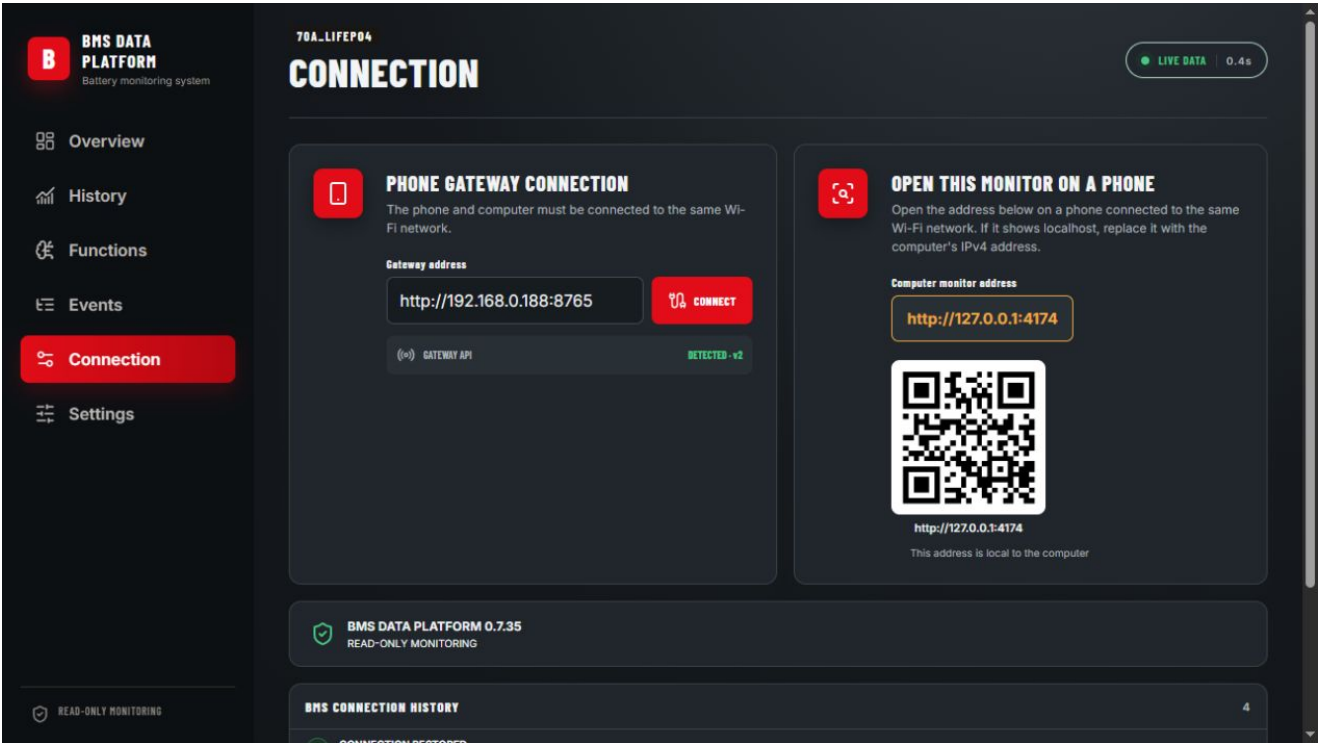
- Connect the phone to the same Wi-Fi network as the computer.
- Enable the local server on the Wi-Fi/gateway page. The app shows a QR code and an address such as `http://192.168.x.x:8765`.
- Do not enter the phone's 127.0.0.1 address on the PC. It points back to the current device. Use the IPv4 address shown by the gateway.
- Keep the app running and remove background-execution restrictions.

Install the Windows monitor

- Download BMS-Data-Platform-0.7.35-Windows-x64.zip from the official release.
- Extract it into a writable folder. Do not run the executable from inside the ZIP file.
- Start BMS Data Platform. If Windows Defender asks, allow access on private networks.
- If EADDRINUSE reports 0.0.0.0:4174, close the already running copy or other software using port 4174.

5.1 Connect the PC to the phone

Enter the gateway address shown by the phone and select Connect. “Gateway API detected · v2” confirms a compatible data contract.



Current Connection page in version 0.7.35.

Gateway address	Phone address, for example <code>http://192.168.0.188:8765</code> . Your network will use different digits.
API v2	Phone and PC use the same data schema. Incompatible versions are blocked with an explicit message.
Monitor address	PC address for opening the interface on another device. <code>127.0.0.1</code> works only on the PC itself; a phone needs the PC's LAN IPv4 address.
QR code	Fast link to the browser monitor. The PC must remain running.

6. Overview page

Overview is the everyday dashboard. Core values fit on one page, and color is an additional cue rather than the only way to communicate state.



Current English overview using real telemetry.

SOC and capacity	BMS state of charge, remaining capacity and nominal capacity. BMS SOC is an estimate and may drift.
Voltage / current / power	Power is calculated as $U \times I$. Positive current and power indicate charging; negative values indicate discharge.
Temperatures	Available battery and MOSFET sensors. The exact set depends on the BMS model.
Imbalance	Maximum active cell voltage minus minimum active cell voltage.
Charge MOS	BMS-reported permission state of the charging transistor; it is not the same as confirmed physical charging current.
Energy flow	Charge: charger → BMS → battery. Discharge: battery → BMS → load.

Time to full or empty

The estimate is smoothed and refreshed less often than instantaneous current to avoid excessive jumping under variable loads. It remains a projection: changing load, inverter efficiency, temperature and usable capacity all change the real runtime.

Cell cards and state colors

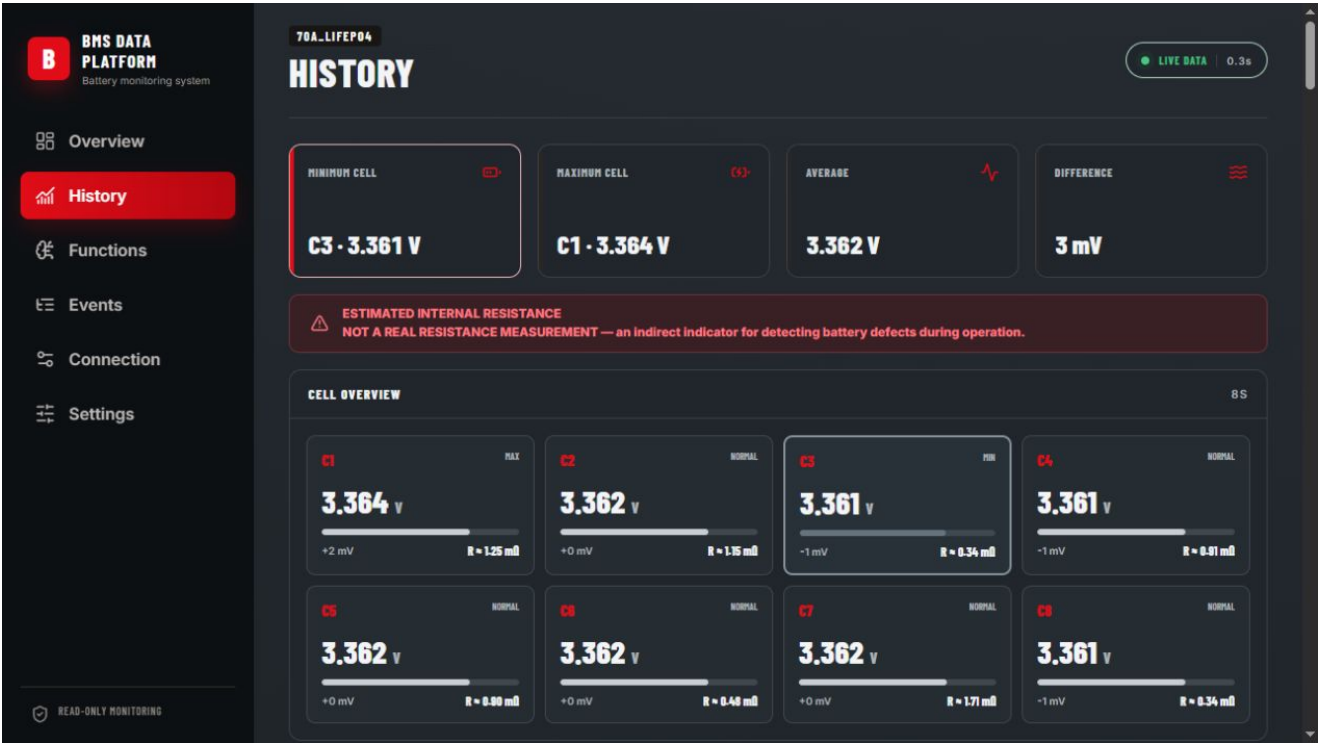
Green / normal	Cell is within the normal range of the selected chemistry and not anomalous relative to the pack.
Yellow / warning	Approaching a monitoring boundary or showing persistent deviation. Investigate the trend; it is not an automatic failure verdict.

Red / critical	Critical boundary, active BMS alarm or strong persistent deviation. Verify actual BMS alarms and current operating conditions.
Balancing	A separate prominent bar/icon identifies a cell currently participating in balancing.
High / low	Text labels make the state understandable without relying on color alone.

SOC disclaimer
Low-cost shunts and BMS coulomb counters may accumulate error. If the battery rarely reaches its calibration point, 17% reported SOC may not equal 17% real energy. LFP voltage is especially flat through the middle of its operating range.

7. History and charts

The PC requests only history that it is missing. A new PC downloads the available phone archive once, then synchronizes new records incrementally.



History page with live cell summary and long-term diagnostic blocks.

Periods	1 hour, 24 hours, 7 days, 30 days and 1 year. Changing the view does not delete data.
Fullscreen	Expands one chart to the whole window. Changing the period must keep the chart in fullscreen.
Zoom	Drag a region with the left button, or in fullscreen hold the right button and use the wheel. Double-click resets zoom.
Time markers	Created independently per chart, draggable and individually hideable.
Gaps	Curves break when telemetry is absent; the application never invents a line across an outage.

7.1 Chart types and controls

Combined history	Multiple parameters in one shared time window with vertically offset scales. Add only the curves needed, choose colors and compare events.
Cell voltages	All active cells on one chart. Each curve can be toggled; Y scaling follows real data and chemistry context.
Current	Bidirectional scale with a zero line: charge above zero, discharge below. Positive and negative colors are configured separately.
Power	Bidirectional for the same reason. The filled shadow points toward the zero line.
Charge / discharge current	Separate positive and negative current charts help when total current visually resembles power.
SOC / temperature / imbalance	Show operating context and allow like-for-like diagnostic comparisons.
Current–power correlation	Scatter plot of power versus current, useful as an overall consistency check.

Thresholds and event markers

- BMS thresholds are read for display only and never written back;
- custom thresholds are PC-local, validated against physical ranges and low < high;
- line color is configured per chart; a small handle on the line moves a custom threshold;
- SOC 0% and 100% flags identify a boundary reach or reset;
- Bluetooth markers show confirmed BMS loss and recovery;
- Charge MOS squares identify permission-state transitions, while actual current remains the evidence of real charging.

8. Cell diagnostics

Natural discharge operating current

A histogram divides discharge current into 1 A bins and measures the share of valid operating time in each bin. The user chooses the maximum current scale and analysis period. The most common bin can serve as a repeatable operating point for future comparisons.

Comparison at equal load

- choose target discharge current and tolerance;
- choose target SOC and tolerance; a solar system may have a typical point well above 50%;
- optionally filter by temperature;
- the monitor keeps one closest snapshot per day;
- cells are compared to the pack median; recurring deviation matters more than one sample.

Balancing behavior

The module evaluates upper charging sessions, per-cell participation, available balancing time and calendar months beginning with the actual first month of statistics. It can reveal cells that repeatedly reach the upper knee first or consume disproportionate balancing effort.

Estimated cell fill and usable capacity

Relative cell fill is learned from accumulated complete cycles, current integration and observed boundaries. A red-to-green gradient highlights limiting versus fuller cells. Treat the result as preliminary until several high-quality cycles have been observed.

Estimated internal resistance

Not a true resistance measurement

The passive indicator estimates $\Delta V/\Delta I$ from operating transitions. Contact resistance, temperature, SOC, electrochemical polarization and packet timing affect it. Use it for trends and persistent anomalies, not as a nameplate DCIR value.

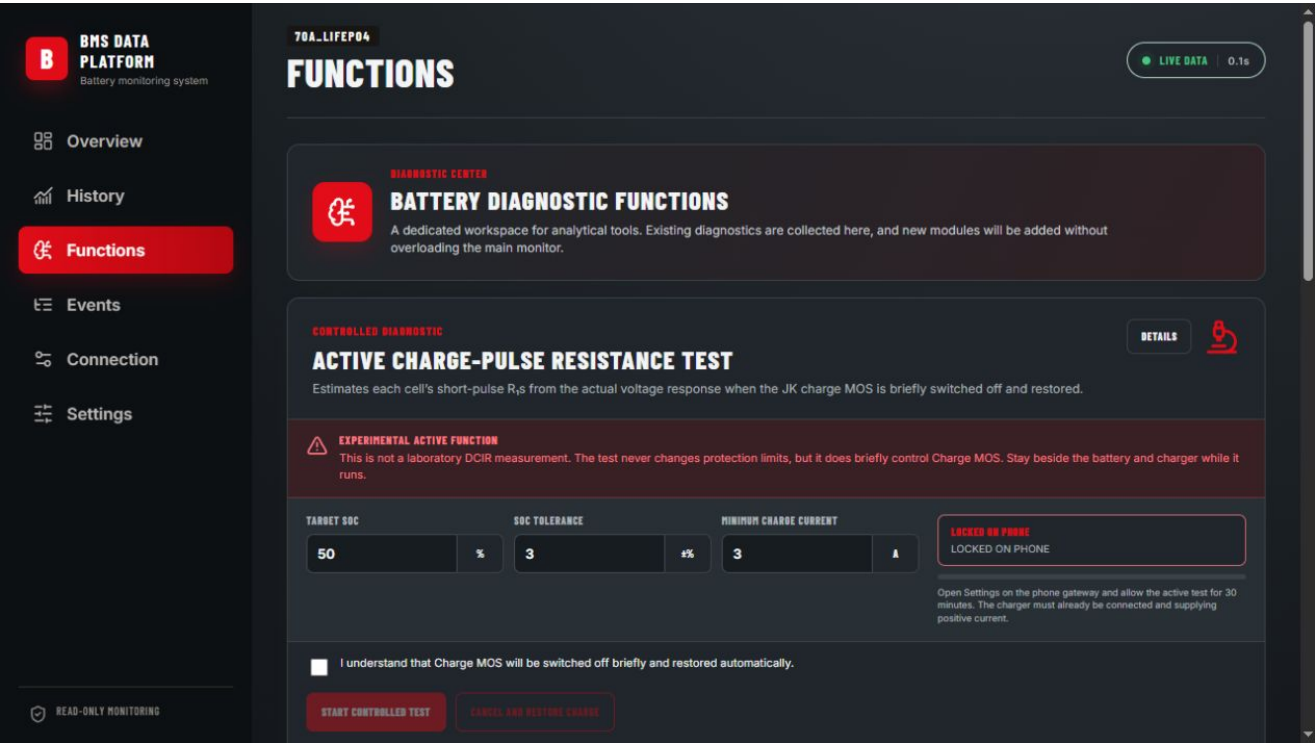
Compare data only at similar current, SOC and temperature. Median processing reduces outliers. A rise relative to the same cell's history and neighboring cells matters more than the absolute milliohm value.

Detecting possible self-discharge

Find a long rest interval with nearly zero current and stable temperature. Compare each cell after the initial relaxation period. A cell that repeatedly loses voltage faster than the pack deserves inspection, but the immediate post-charge drop can be normal relaxation.

9. Functions page

This page separates daily monitoring from advanced diagnostic tools. “Available” means the algorithm exists; it does not mean sufficient data have already been collected.



Diagnostic center and experimental active pulse test.

Available	Module is implemented. A result may still require history, complete cycles or specific operating conditions.
Planned	Describes future work and does not perform analysis in the current release.
Advanced diagnostics	Reveals technical controls and methodology for experienced users.
GATT status codes	Adds Android GATT error codes to confirmed outages. Useful for troubleshooting, optional for normal use.

9.1 Active pulse-resistance test

Experimental active function

The test briefly disables Charge MOS. Run it only while physically present near the battery and charger. It is not a laboratory DCIR test and never changes BMS protection thresholds.

Prerequisites

- the phone has locally armed the test for 30 minutes; permanent authorization intentionally does not exist;
- phone and PC use compatible API versions;
- the charger is physically connected and current is positive and above the chosen minimum;
- SOC is within the target window, while current and communication are stable;
- the operator has explicitly acknowledged the Charge MOS transition.

Measurement sequence

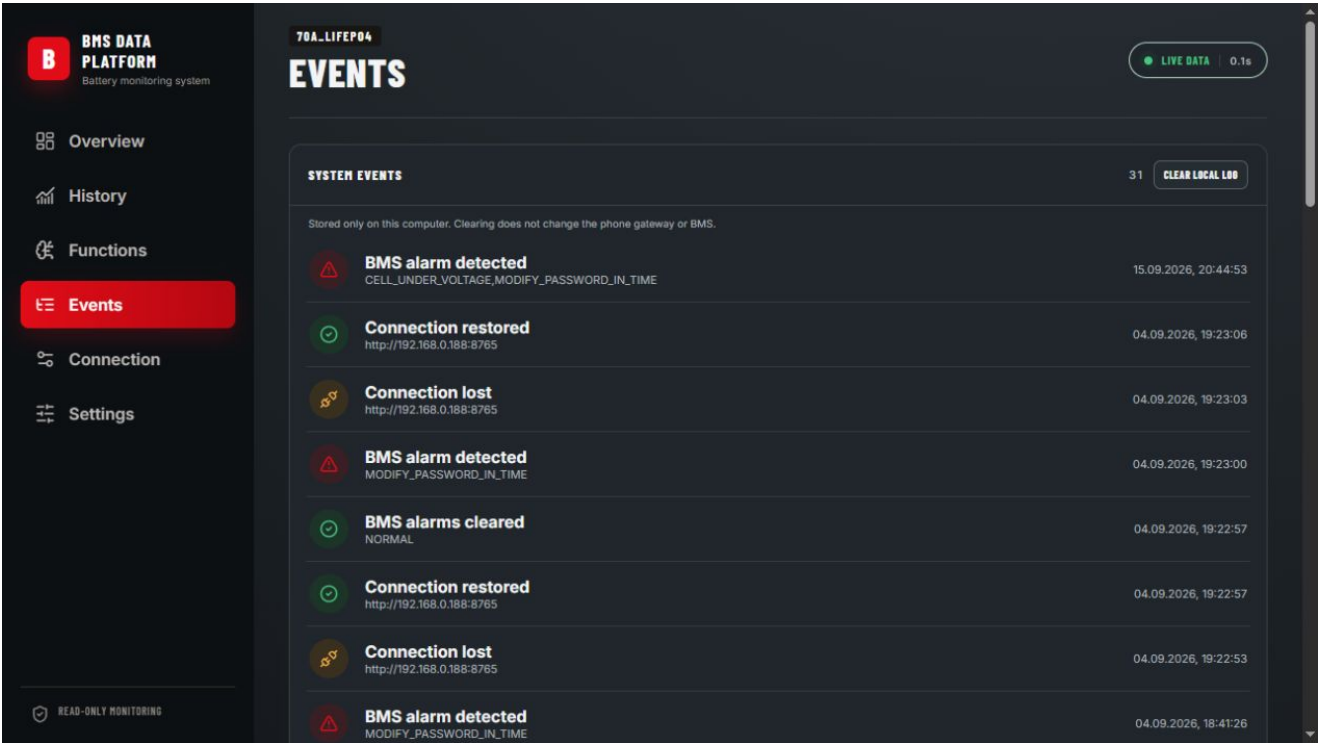
- collect a stable baseline of current and every cell voltage during charging;
- command Charge MOS off and verify that current actually changes;
- measure the fast voltage drop after charge current removal, separating it from slow relaxation;
- automatically restore Charge MOS to its original state;
- measure the reverse edge after current returns;
- compare off-edge and on-edge estimates and store quality information for every cell;
- on cancellation or error, restoring the charge path is the first priority.

Where results are shown

Results are stored on the phone and displayed under "Pulse test history and comparison". Repeat the test months later at similar SOC, temperature and charge current. Compare each cell against its own history and the pack median.

10. Events and alarms

Events reconstruct what happened: connection loss, restoration, BMS alarms, Charge MOS transitions, SOC boundaries and completed charging sessions.



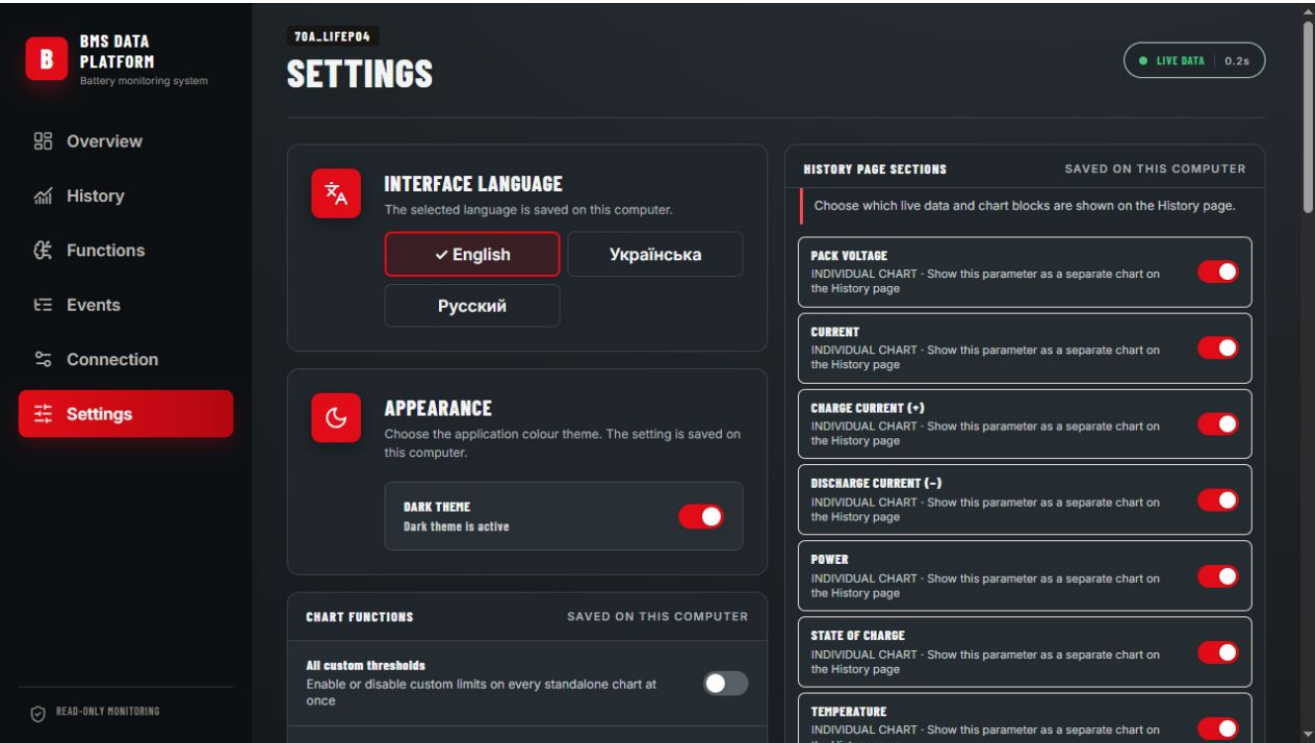
System event log and saved charging sessions.

System events	Stored on the current PC. “Clear on this PC” does not erase phone history or change the BMS.
BMS alarm	A red card includes decoded alarm bits. Alarm clearance creates a separate event.
0x80000	Modify password in time is a password-change reminder, not an electrical battery fault.
Connection lost / restored	Confirmed outages include time and duration. Short packet errors should not generate false outage events.
Charged this session	Integral of positive BMS current over one session, with peak current and timestamps. Charge MOS off ends the active session but keeps the record.

Charge MOS state is not proof of actual current
A BMS may report charge disabled while a bench supply or circuit still produces observed charging current. Always compare MOS state with the measured current sign and magnitude. Treat a contradiction as a diagnostic event.

11. Settings

PC settings never rewrite the BMS. Display preferences are local, allowing a simple interface for everyday users and a detailed one for diagnostics.



Language, theme, chart, threshold and export settings.

Language and theme	English, Ukrainian, Russian; light and dark themes.
History blocks	Each chart and diagnostic block can be enabled separately. The crossed-eye button hides a card directly from History.
BMS thresholds	Decoded read-only values with explanations. Availability depends on model and protocol.
Custom thresholds	Local visual guides, never sent to the BMS. Invalid physical values and $low \geq high$ are rejected.
Curves and colors	Cell curves and positive/negative current and power colors are configured independently.
Symmetric scale	Useful for charge/discharge comparison; auto range uses space better when all samples have one sign.

12. History synchronization, export and backup

Incremental synchronization

- every record carries a date and time; the phone is the primary recorder while attached to the BMS;
- the PC remembers the newest received record and requests only the missing interval next time;
- a fresh PC has no local cache, so its first synchronization downloads the entire available phone archive;
- if the phone is absent, already synchronized PC data remain available but cannot be presented as live;
- a large first archive can take time; later incremental updates are much shorter.

SQL/SQLite export

In Settings choose “Download SQL database”. Export should work from already cached PC history even while the phone is unavailable. The file contains telemetry, cell voltages, temperatures, SOC events, connections, charge sessions and synchronization metadata. Open a copy in DB Browser for SQLite; do not edit the application's live database.

Excel export

“Export to Excel” creates separate sheets for telemetry, events, connections and export metadata. Excel is convenient for filtering and user charts; SQL is preferable for backup and automated processing.

Backup and migration

- create an SQL backup before every major update;
- connect the same phone to a new PC to rebuild the available local history and continue incrementally;
- for archives longer than the phone retention period, keep periodic SQL exports;
- never merge databases manually without a backup; use battery identity and timestamps to prevent duplicates.

13. Local API and external integrations

The Android gateway exposes its current snapshot as JSON. The primary integration endpoint is:

```
GET http://<PHONE_IP>:8765/api/monitoring/v1/snapshot
```

- use the phone address shown on the Wi-Fi gateway page;
- check timestamp/data age and connection state before consuming values;
- do not interpret a missing field as zero because BMS models expose different sensors;
- for Zabbix, use an HTTP agent/master item and JSONPath dependent items;
- recommended priorities are cell voltages, SOC, temperatures, alarms, balancing, pack voltage and current;
- poll at a reasonable rate and never expose the endpoint directly to the Internet.

API compatibility

The PC and phone exchange a compatibility identifier. An incompatible pair is blocked entirely so an old client cannot misinterpret a new schema. Install both components from the same release.

14. Updating and troubleshooting

BMS not found	Enable Bluetooth and, on Android 8–11, Location. Grant permissions, close the JK app, move closer and wait for the full 25-second scan.
Repeated BMS beeps	Stop repeated attempts, close competing apps, cycle Bluetooth once and perform one scan. Confirm that reconnect targets the selected device.
Phone connected, PC empty	Check same Wi-Fi, the :8765 address, API v2 status, firewall access and fresh BMS telemetry on the phone.
History keeps loading	The first sync may transfer a large archive. Avoid restarting repeatedly. Later syncs transfer only the difference.
White browser page	Update the browser. Older browsers may lack crypto.randomUUID; the current release uses a compatibility fallback. Clear site data only after preserving required local history.
EADDRINUSE :4174	Another instance owns the port. Close it in Task Manager and start one copy.
Incompatible version	Update Android and Windows together from the same release. Data exchange is deliberately blocked until API versions match.
Chart not updating	Check freshness, selected time range, hidden-block state, connection gaps and synchronization completion.
Alarm remains visible	Current alarm state and the event log are different. A cleared BMS adds a clearance event; the historical event remains.
SOC looks wrong	Check zero-current calibration, nominal capacity and the last full calibration. BMS SOC can drift.

Safe update sequence

- create an SQL backup;
- install the new APK over the old one without uninstalling;
- extract the new Windows build to a new folder;
- run one instance and verify API v2, live data and history synchronization;
- archive the old folder only after verification, and keep database backups.

Glossary and operating checklist

SOC	Estimated state of charge; not a direct energy measurement.
Imbalance	Highest active cell voltage minus lowest active cell voltage.
Charge / Discharge MOS	Power transistors that permit charging and discharging.
Balancing	Energy redistribution/dissipation used to bring cell voltages closer.
DCR / DCIR	DC internal resistance, dependent on method, SOC and temperature.
Overpotential	Current- and polarization-related voltage rise during charge.
Underpotential / sag	Measured voltage depression under load.
GATT	Android BLE layer; an error code can help explain an outage.
Stale data	The last known snapshot that must no longer be treated as current.

Daily check

- data status is live and packet age is small;
- pack voltage agrees with the cell sum;
- current sign matches the real operating mode;
- there are no new alarms or unexpected MOS transitions;
- imbalance and temperature do not show a persistent worsening trend;
- the phone is recording history and remains reachable over Wi-Fi.

Support and downloads

Official release: <https://github.com/zayants/bms-data-platform/releases/tag/v0.7.35>

Android APK: <https://github.com/zayants/bms-data-platform/releases/download/v0.7.35/BMS-Gateway-0.4.9-Android.apk>

Windows x64:

<https://github.com/zayants/bms-data-platform/releases/download/v0.7.35/BMS-Data-Platform-0.7.35-Windows-x64.zip>